The Canadian Urban Environmental Health Research Consortium

Canue Metadata - Greenness Tree Canopy Coverage

2026-11-06

DATA SET INFORMATION

Dataset Code: GRTCC_AVA_YY

Description:

Tree canopy is defined as area of vegetation (including leaves, stems, branches, etc.) of woody plants above 5m in height. The dataset developers derived tree canopy cover estimates from the Global Forest Cover Change (GFCC) Surface Reflectance product (GFCC30SR), which is based on enhanced Global Land Survey (GLS) datasets. The GLS datasets are composed of high-resolution Landsat 5 Thematic Mapper (TM) and Landsat 7 Enhanced Thematic Mapper Plus (ETM) images at 30 meter resolution.CANUE staff retrieved tree canopy cover data from Google Earth Engine (GEE) for the year 2010 and 2015, extracted values (percent coverage) to postal codes and calculated summary measures (average percent coverage) within buffers of 100, 250, 500, and 1000 metres.

Keywords: tree canopy|forest|greenness|landsat

Place Keywords: national|Canada

GEOSPATIAL REFERENCE

Upper Left Corner: 65.14N , -141.02W

Lower Right Corner: 41.68N , -52.62W

Coordinate System: GCS_WGS84 - EPSG:4326

Geometry Type: POINT - Units: Decimal Degree

Geometry Data Source: DMTI Spatial Inc. (postal codes)

QUALITY ASSESSMENT

QA/QC Procedures:

CANUE did not assess the quality of the tree canopy data. Users should review the supporting documentation and any recommended citations.

Geographic Coordinate Positional Accuracy:

These metrics are linked to the corresponding annual postal codes files for mapping and analysis purposes. Refer to the postal code metadata file in Supporting Documentation for more information.

Vertical Positional Accuracy: N/A

Attribute Accuracy: N/A

Data Validity: NoData = -9999 (for numeric fields) - NoData=null (for category fields) - Data insufficient to calculate value = -1111

Associated Files: N/A

Data Comment:

DATA SOURCE

Data Source

Data were accessed via Google Earth Engine and are described in the User Guide (see links in supporting documentation).

Spatial Resolution: 30 metres

Data Preparation Date: 2021-11-24

Beginning Date: 2010

End Date: 2015

Sampling Frequency of Data: Annual

Years Available:
2010 - 2015

MAINTENANCE

Description:

File Type: Comma separated values(.csv)

File Size:

Number of Data Files: 2

DATA USE CONDITIONS

The Data User is REQUIRED:
(i) to acknowledge data sources listed under Acknowledgement(s)
(ii) cite the publication(s) listed under Recommended Citation(s) as the providers and source of these data when using them in support of research, analysis, operations, policy decision or any other undertaking including publication
(iii) complete and sign the CANUE Data Use and Sharing Agreement in which the name and signature of the researcher/analyst who takes responsibility for ensuring all conditions are met.

Data Sharing Restrictions:
These data files are provided solely for the purposes stated in the CANUE Data Sharing and Use Agreement and should not be re-distributed for any reason. These data also contain proprietary postal code data and may only be used for the project named in the CANUE Data Sharing and Use Agreement. Data can be shared only within a project team for the exclusive purposes of teaching, academic research and publishing, and/or planning of educational services in accordance to DMTI End User Agreement associated with the Spatial Mapping Academic Research Tools (SMART) Program.

Include the following references in any publications resulting from the use of these data:
[1]Sexton, J. O., Song, X.-P., Feng, M., Noojipady, P., Anand, A., Huang, C., Kim, D.-H., Collins, K.M., Channan, S., DiMiceli, C., Townshend, J.R.G. (2013). Global, 30-m resolution continuous fields of tree cover: Landsat-based rescaling of MODIS Vegetation Continuous Fields with lidar-based estimates of error. International Journal of Digital Earth, 130321031236007.
[2] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., and Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. Remote Sensing of Environment.
[3]CanMap Postal Code Suite v2010 and v2015. [computer file] Markham: DMTI Spatial Inc., 2010 and 2015.

Include the following acknowledgements:
1. Tree Canopy metrics, indexed to DMTI Spatial Inc. postal codes , were provided by CANUE (Canadian Urban Environmental Health Research Consortium).
2. Intellectual property rights to this dataset belong to University of Maryland, Department of Geographical Sciences and NASA.

SUPPORT DOCUMENTATION

1 - GFCC User Guide (https://canuedata.ca/docs/GFCC_User_Guide_V1.pdf)
2 - Global 30-m resolution continuous fields of tree cover: Landsat-based rescaling of MODIS vegetation continuous fields with lidar-based estimates of error. (https://canuedata.ca/docs/Global_30m_Tree_Cover.pdf)
3 - Earth Science Data Records of Global Forest Cover and Change - Algorithm Theoretical Basis Document (https://canuedata.ca/docs/GFCC_ATBD.pdf)
4 - Postal Code metadata (https://canuedata.ca/docs/CANUE-Browser-Metadata-PostalCodes.pdf)

VARIABLES

GRTCCYY_01 - Annual Value at Postal Code

Estimated percent of pixel covered by vegetation greater than 5m in height

GRTCCYY_02 - Annual Average 100m

Average of estimated percent of pixel covered by vegetation greater than 5m in height within 100m

GRTCCYY_03 - Annual Average 250m

Average of estimated percent of pixel covered by vegetation greater than 5m in height within 250m

GRTCCYY_04 - Annual Average 500m

Average of estimated percent of pixel covered by vegetation greater than 5m in height within 500m

GRTCCYY_05 - Annual Average 1000m

Average of estimated percent of pixel covered by vegetation greater than 5m in height within 1 km

SUPPORT CONTACT

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